

A report on

**Sliding Mode Control for Trajectory Tracking of a
Quadrotor under Single Rotor Fault**

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2026

Abstract

Quadrotor unmanned aerial vehicles (UAVs) are widely used in aerial robotics due to their agility and mechanical simplicity. However, their stability relies on the coordinated operation of four independent rotors. A fault in one rotor, such as loss of effectiveness (LOE) or complete failure, significantly degrades the controllability of the system and may lead to instability during flight. This report investigates a sliding mode control (SMC) strategy for trajectory tracking of a six degree-of-freedom (6-DOF) quadrotor in the presence of a single rotor fault. The objective is to design a robust control law that enables the quadrotor to follow a desired trajectory despite the loss of thrust effectiveness in one actuator. A control framework based on the nonlinear dynamics of the quadrotor will be developed using sliding mode control to ensure robust trajectory tracking. The proposed controller will be evaluated through simulations of the nonlinear quadrotor model under rotor fault conditions. The results will illustrate the capability of the SMC-based controller to maintain stable flight and achieve satisfactory trajectory tracking performance despite actuator degradation.

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Chapter 1

Introduction

Unmanned aerial vehicles (UAVs), particularly quadrotor systems, have attracted significant attention in recent years due to their simple mechanical structure, vertical take-off and landing capability, and high maneuverability. These characteristics make quadrotors suitable for a wide range of applications such as surveillance, inspection, environmental monitoring, search and rescue, and logistics operations. Despite their advantages, quadrotors are inherently nonlinear, underactuated, and strongly coupled dynamical systems, which makes control design a challenging task. Achieving stable flight and accurate trajectory tracking requires the coordinated operation of four independent rotors that generate the thrust and torques necessary to control the vehicle motion.

Despite these advantages, quadrotors remain highly sensitive to actuator faults because their flight stability relies heavily on the coordinated operation of all four rotors. Rotor faults are particularly critical because they directly affect the thrust generation mechanism of the vehicle. A loss of effectiveness (LOE) in rotor or complete rotor failure significantly degrades the controllability of the system and may lead to instability during flight. In particular, the loss of a single rotor prevents independent control of roll, pitch, yaw, and thrust simultaneously, which may lead to instability and ultimately a crash if appropriate fault-tolerant control strategies are not employed. Research has shown that stable flight may still be achievable after rotor failure by sacrificing control over certain states, such as the yaw angle, and allowing the vehicle to spin around its vertical axis while maintaining control of the remaining critical states [1].

Several studies have investigated the behavior and controllability of quadrotors under rotor failure conditions. Mueller and D'Andrea has demonstrated that even after the complete loss of one or more propellers, a quadrotor can maintain controlled motion by spinning about a fixed axis and adjusting thrust to regulate position [2]. By tilting this rotational axis, translational motion can still be achieved, enabling the vehicle to maintain altitude and limited maneuverability despite actuator loss. These results highlight the possibility of achieving degraded but controllable flight under severe actuator faults.

In addition to actuator failures, quadrotor systems are often affected by modeling uncertainties, external disturbances, and parameter variations, all of which can degrade

control performance. To address uncertainties and disturbances in nonlinear systems, sliding mode control (SMC) has been widely used due to its robustness properties and strong disturbance rejection capability. Sliding mode control forces the system trajectories to evolve on a predefined sliding surface, thereby ensuring robustness with respect to model uncertainties and external disturbances. In recent years, sliding mode based fault-tolerant control strategies have been explored for quadrotor systems. For example, Zhao *et al.* proposed a fault observer based adaptive sliding mode control strategy for quadrotors subject to actuator faults and disturbances, demonstrating improved tracking performance and robustness [3].

Motivated by existing research, this project investigates the design of a sliding mode control strategy for trajectory tracking of a quadrotor under a single rotor fault. The objective is to develop a control framework capable of maintaining stable flight and tracking desired trajectories despite the loss of one actuator. The proposed approach leverages the robustness properties of SMC to compensate for the underactuated dynamics introduced by rotor failure while ensuring accurate tracking performance.

The structure of this report is organized as follows. First, the dynamic model of the quadrotor system is presented, followed by the control allocation and way to induce the fault in the dynamics. Next, the problem formulation is described.

Chapter 2

System Description

2.1 System Description

The system considered in this project is a quadrotor unmanned aerial vehicle modeled as a rigid body with six degrees of freedom (6-DOF). The quadrotor consists of four rotors mounted at the ends of a cross-shaped frame, each producing an upward thrust force and an associated reaction torque. The vehicle motion is described using two coordinate frames: an inertial frame $\mathcal{I}$ fixed to the earth and a body-fixed frame $\mathcal{B}$ attached to the center of mass of the quadrotor.

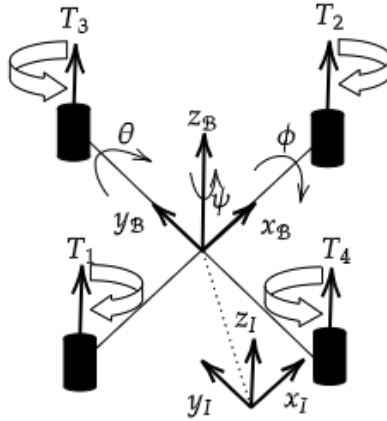


Figure 2.1: Configuration of Quadrotor

The state of the quadrotor is represented by the vector

$$x = [x, y, z, \phi, \theta, \psi, \dot{x}, \dot{y}, \dot{z}, \dot{\phi}, \dot{\theta}, \dot{\psi}]^T$$

where (x, y, z) denote the position of the quadrotor in the inertial frame, (ϕ, θ, ψ) represent the roll, pitch, and yaw angles respectively, and the remaining states represent the corresponding linear and angular velocities.

Integral error states are also included in the simulation model to support integral action in the controller.

2.1.1 Quadrotor Dynamic Model

The translational dynamics of the quadrotor are derived from Newton's second law and are expressed as

$$m\ddot{p} = Rf_T - mg$$

where m is the mass of the quadrotor, $p = [x, y, z]^T$ is the position vector, R is the rotation matrix from body frame to inertial frame, f_T is the total thrust generated by the four rotors, and g is gravitational acceleration.

Expanding the translational equations yields

$$\ddot{x} = \frac{T}{m}(\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \quad (2.1)$$

$$\ddot{y} = \frac{T}{m}(\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \quad (2.2)$$

$$\ddot{z} = \frac{T}{m}(\cos \phi \cos \theta) - g \quad (2.3)$$

where T represents the total thrust generated by the rotors.

The rotational dynamics are described using Euler's rigid body equations

$$I\dot{\omega} + \omega \times (I\omega) = \tau$$

where $I = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ is the inertia matrix, $\omega = [\dot{\phi}, \dot{\theta}, \dot{\psi}]^T$ is the angular velocity vector, and $\tau = [\tau_\phi, \tau_\theta, \tau_\psi]^T$ represents the control torques.

The corresponding rotational acceleration equations are

$$\ddot{\phi} = \frac{I_{yy} - I_{zz}}{I_{xx}}\dot{\theta}\dot{\psi} + \frac{\tau_\phi}{I_{xx}} \quad (2.4)$$

$$\ddot{\theta} = \frac{I_{zz} - I_{xx}}{I_{yy}}\dot{\phi}\dot{\psi} + \frac{\tau_\theta}{I_{yy}} \quad (2.5)$$

$$\ddot{\psi} = \frac{I_{xx} - I_{yy}}{I_{zz}}\dot{\phi}\dot{\theta} + \frac{\tau_\psi}{I_{zz}} \quad (2.6)$$

2.1.2 Control Inputs and Allocation

The control inputs to the system are defined as

$$U = [u_1, u_2, u_3, u_4]^T$$

where

$$u_1 = T$$

is the total thrust, and

$$u_2 = \tau_\phi, \quad u_3 = \tau_\theta, \quad u_4 = \tau_\psi$$

represent the roll, pitch, and yaw torques respectively.

These inputs are generated by the individual rotor thrusts T_1, T_2, T_3, T_4 . The relationship between rotor thrusts and control inputs is expressed through the control allocation matrix

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} \\ \frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} \\ d & -d & d & -d \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} \quad (2.7)$$

where l denotes the arm length and d is the torque-to-thrust coefficient.

2.1.3 Rotor Fault Model

To model actuator faults, the thrust generated by rotor 1 is scaled by an effectiveness parameter α . The fault model is given by

$$T_1^{fault} = \alpha T_1 \quad (2.8)$$

where $0 \leq \alpha \leq 1$. A value $\alpha = 1$ represents normal operation, while $\alpha = 0$ corresponds to complete rotor failure.

When rotor 1 fails completely, the system operates using the remaining three rotors and the control allocation matrix is reduced accordingly.

Using this dynamic model, the objective of the control system is to enable the quadrotor to track a desired trajectory while maintaining stability even in the presence of actuator faults.

Table 2.1: Notation used in the quadrotor dynamic model

Symbol	Description
x, y, z	Position of the quadrotor in the inertial frame (m)
ϕ, θ, ψ	Roll, pitch, and yaw angles (rad)
$\dot{x}, \dot{y}, \dot{z}$	Linear velocities (m/s)
$\dot{\phi}, \dot{\theta}, \dot{\psi}$	Angular velocities (rad/s)
m	Mass of the quadrotor (kg)
g	Gravitational acceleration (m/s ²)
I_{xx}, I_{yy}, I_{zz}	Moments of inertia (kg m ²)
p	Position vector $[x \ y \ z]^T$
R	Rotation matrix from body to inertial frame
ω	Angular velocity vector
T	Total thrust (N)
T_1, T_2, T_3, T_4	Individual rotor thrusts (N)
l	Arm length (m)
d	Torque-to-thrust coefficient
$\tau_\phi, \tau_\theta, \tau_\psi$	Roll, pitch, yaw torques (Nm)
u_1	Total thrust control input
u_2, u_3, u_4	Roll, pitch, yaw torque inputs
U	Control input vector
A	Control allocation matrix
α	Rotor effectiveness factor
T_1^{fault}	Effective thrust of rotor 1 under fault

Chapter 3

Problem Statement

The quadrotor is a nonlinear, underactuated aerial vehicle with six degrees of freedom (three translational and three rotational) and four control inputs generated by the rotors. The translational motion of the vehicle determines its position in the inertial frame, while the rotational motion determines its orientation described by roll (ϕ), pitch (θ), and yaw (ψ) angles.

The dynamic model of the quadrotor, derived using the Newton–Euler formulation in the previous section, can be expressed in the general nonlinear form

$$\dot{x} = f(x) + g(x)u$$

where $x \in \mathbb{R}^n$ represents the system state vector consisting of position, velocity, attitude angles, and angular velocities, $u \in \mathbb{R}^4$ denotes the control inputs corresponding to the collective thrust and control torques.

In practical flight scenarios, actuator faults may occur due to motor degradation, propeller damage, or electronic failures. A common type of actuator fault is the loss of effectiveness (LOE) in a rotor, where the generated thrust becomes smaller than the commanded value. In the extreme case, the rotor may completely fail and produce no thrust. Such faults directly affect the control authority of the quadrotor and can severely degrade its stability and trajectory tracking performance.

In this work, a single rotor fault is considered. The fault is modeled as a reduction in the effectiveness of the thrust generated by one rotor. If T_i denotes the nominal thrust produced by the i^{th} rotor, the faulty thrust can be expressed as

$$T_i^{\text{fault}} = \alpha_i T_i$$

where $0 \leq \alpha_i \leq 1$ represents the rotor effectiveness factor. A value $\alpha_i = 1$ corresponds to normal operation, $0 < \alpha_i < 1$ represents partial loss of effectiveness, and $\alpha_i = 0$ corresponds to complete rotor failure.

The presence of such actuator faults alters the input distribution of the quadrotor and introduces additional uncertainty into the control system. Consequently, the controller

must maintain stability and tracking performance despite this degraded actuation. The primary control objectives are:

- Stabilization of the quadrotor with the presence of fault in rotor.
- Tracking of desired trajectories.
- Maintaining robustness against model uncertainties and disturbances.

To achieve these objectives, a sliding mode control strategy will be developed to provide robustness against model uncertainties, disturbances, and actuator degradation. The controller will be designed based on the nonlinear quadrotor dynamics presented earlier, and its effectiveness will be evaluated through numerical simulations of trajectory tracking under rotor fault conditions.

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